Ring Oscillator Design and Verification in Cadence Virtuoso

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I. Introduction

Ring oscillators are inverter-based circuits that convert a DC voltage into an AC signal. The transient ripple caused by the start-up voltage is amplified as it passes through inverters in a loop; the time it takes to complete one loop is half the period and sets the frequency and duty cycle. The delays between inverter stages depend on both the loaded capacitance and the capacitance of each stage due to the gate area and channel width. The duty cycle is defined as the time from a rise to the next fall divided by the time from that rise to the next rise. The frequency is the inverse of the period, the time between one rise and the next. Well-designed ring oscillators can produce precise, continuous square waves for clocking when integrated with phase locked loops. In this lab a ring oscillator is designed using GPDK045 on Cadence Virtuoso using the design targets found in Table I.

TABLE I: Design Targets

	Parameter	Value
Requirements	Frequency f	1 GHz
	Devices	nmos1v / pmos1v (standard V_T)
	Duty cycle D	50%
Goals	Supply V_{DD}	1.1 V
	Channel length L	45 nm
	Width grid / minimum	5 nm / 120 nm
	Compared stage counts n	3, 5, 7, 19, 51

II. Design Equations

The oscillator's frequency and duty cycle depend on the propagation delays tpHL and tpLH of the inverters. These are dependent on both the operating conditions and device parameters, each of which can be tuned or calculated before layout design using:

$$t_{pHL} = \frac{2C_L V_{DD}}{k'_n (W_n/L_n)(V_{DD} - V_{Tn})^2}, \qquad t_{pLH} = \frac{2C_L V_{DD}}{k'_p (W_p/L_p)(V_{DD} - |V_{Tp}|)^2} \tag{1}$$

assuming the switching transistor carries its full saturation current for the whole transition, and the swing is close to the source voltage.

The rise-and-fall spacing depends heavily on the capacitance between stages. This is called the loading capacitance, which depends on the gate oxide area, channel width, and external capacitance. Since external capacitance is easiest to control, the desired loaded capacitance is calculated from a rearrangement Eq. 1 and the collective results of Eq. 3 and Eq. 4 using:

$$C_L = C_{ox}(W_n L_n + W_p L_p) + C_{jn} W_n + C_{jp} W_p + C_{EXT} \tag{2}$$

$$f = \frac{1}{n\,(t_{pHL} + t_{pLH})} \tag{3}$$

$$D = \frac{(n+1)\,t_{pHL} + (n-1)\,t_{pLH}}{2n\,(t_{pHL} + t_{pLH})} = \frac{1}{2} + \frac{r-1}{2n\,(r+1)}, \qquad r = \frac{t_{pHL}}{t_{pLH}} \tag{4}$$

Because process design kits (such as gpdk045) often restrict device dimensions and operating conditions, the best way to meet a design goal is to minimize error while respecting those limits. This was done with Python optimization scripts using equations for channel width and external capacitance,

$$\frac{W_p}{W_n} = r\,\frac{k'_n (V_{DD} - V_{Tn})^2 L_p}{k'_p (V_{DD} - |V_{Tp}|)^2 L_n} \tag{5}$$

$$C_{EXT} = \frac{C_{dev}}{nf\,(t_{pHL,0} + t_{pLH,0})} - C_{dev}, \qquad C_{dev} = C_L\big|_{C_{EXT}=0} \tag{6}$$

while considering the minimum channel width (120nm), step size (5nm), and realistic capacitance values when it can (Eq. 6 $>$ 0.5 fF for $n \leq 29$). The optimization code can be found in this GitHub repository: **Ring Oscillator**.

III. Device Parameters

The parameters for nmos1v and pmos1v in gpdk045 were found online at **MS19 45nm PDK predictive model development project undertaken for Cadence**, derived (k') or assumed (TOXE) if not listed, and are presented below in Table II.

TABLE II: GPDK045 Device Parameters

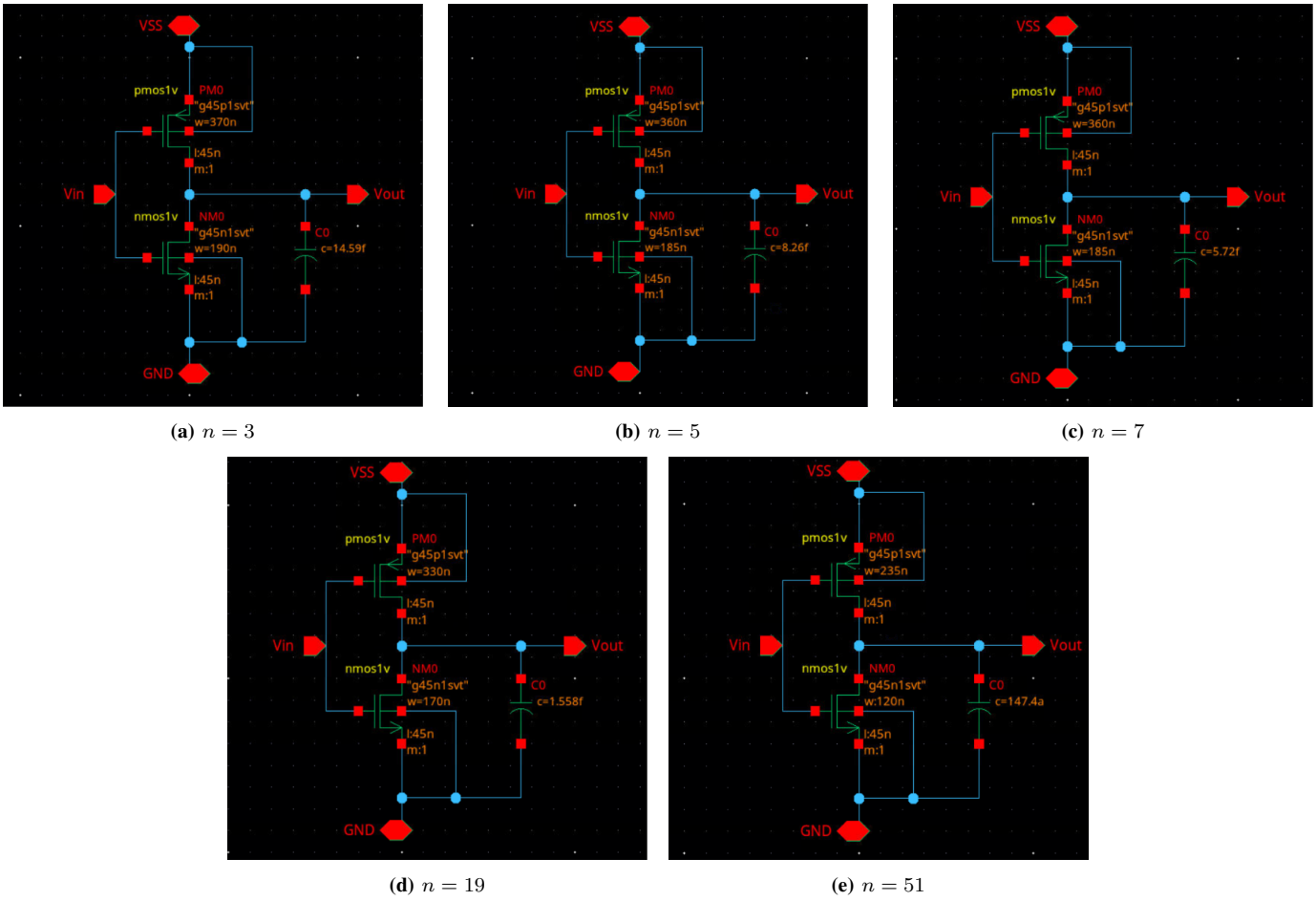
Parameter	NMOS (nmos1v)	PMOS (pmos1v)
V_{th} (V)	0.38	-0.43
I_{dsat} at V_{DD} (mA)	5.96	3.06
k' (µA/V^2)	92.0	54.5
C_J (F/m^2)	1.177×10^{-3}	1.308×10^{-3}
C_{JSW} (F/m)	6.562×10^{-11}	7.604×10^{-11}
C_j per width (fF/µm)	0.296	0.335
C_{ox} (fF/µm^2)	19.2	19.2

Using parameters found online, the Python script *design_optimization.py* sets Wp/Wn from Eq. 5, picks the smallest Wn on the 5 nm grid whose rounded Wp keeps $|D - 0.5| < 0.01\%$, then solves C_{EXT} exactly from Eq. 6 while fitting PDK constraints. To compare the number of inverters and the transient behavior of a ring oscillator, 5 configurations were simulated and are highlighted below in Table III.

TABLE III: Calculated Designs for All Odd n ($V_{DD} = 1.1\,\mathrm{V}$, $L = 45\,\mathrm{nm}$)
Highlighted Rows Simulated in Cadence

n	W_n (nm)	W_p (nm)	C_{EXT} (fF)	f (GHz)	D	f err (%)	D err (%)	P (µW)
3	190	370	14.5861	1.000002	0.499985	0.0002	–0.0029	55.36
5	185	360	8.2599	1.000005	0.499955	0.0005	–0.0091	53.88
7	185	360	5.7154	1.000002	0.499968	0.0002	–0.0065	53.88
9	180	350	4.1840	0.999997	0.499953	–0.0003	–0.0093	52.40
11	180	350	3.3091	0.999988	0.499962	–0.0012	–0.0076	52.40
13	175	340	2.6272	1.000015	0.499952	0.0015	–0.0096	50.93
15	175	340	2.1956	0.999989	0.499958	–0.0011	–0.0083	50.93
17	170	330	1.8115	0.999984	0.499951	–0.0016	–0.0099	49.45
19	170	330	1.5584	1.000003	0.499956	0.0003	–0.0088	49.45
21	170	330	1.3536	0.999975	0.499960	–0.0025	–0.0080	49.45
23	165	320	1.1490	1.000028	0.499953	0.0028	–0.0093	47.97
25	165	320	1.0111	1.000026	0.499957	0.0026	–0.0086	47.97
27	160	310	0.8663	0.999965	0.499951	–0.0035	–0.0097	46.50
29	120	235	0.5784	1.000003	0.500047	0.0003	0.0094	35.06
31	120	235	0.5139	1.000044	0.500044	0.0044	0.0088	35.06
33	120	235	0.4573	0.999992	0.500041	–0.0008	0.0082	35.06
35	120	235	0.4071	1.000023	0.500039	0.0023	0.0078	35.06
37	120	235	0.3624	0.999961	0.500037	–0.0039	0.0073	35.06
39	120	235	0.3222	1.000012	0.500035	0.0012	0.0070	35.06
41	120	235	0.2860	0.999953	0.500033	–0.0047	0.0066	35.06
43	120	235	0.2531	0.999995	0.500032	–0.0005	0.0063	35.06
45	120	235	0.2231	1.000074	0.500030	0.0074	0.0060	35.06
47	120	235	0.1957	1.000077	0.500029	0.0077	0.0058	35.06
49	120	235	0.1706	0.999973	0.500028	–0.0027	0.0055	35.06
51	120	235	0.1474	0.999990	0.500027	–0.0010	0.0053	35.06

After optimizing device values for the inverter chains, schematics for 5 designs were created, these are shown below in Fig. 1. Symbols were then created to include source voltage and GND pins on both the left and right sides, with Vin on the left and Vout on the right. When implemented in series the voltage sources and grounds are tied between inverters and the output of the first connects directly to the input of the next without any additional wiring. Instance arrays could then be created and spaced such that arrays of any N could be joined with 2 clicks and a wire connection between Vout of the last and Vin of the first. To avoid errors from floating pins, the source of the last must be tied to the first, and the final ground pin must be grounded. Fig. 2 shows a close-up of the input side of 4 chains; I omitted a full image of the ring oscillators because capturing its width would lose most of the important information due to the width of the N=51 chain.



(a) $n = 3$ **(b)** $n = 5$ **(c)** $n = 7$

(d) $n = 19$ **(e)** $n = 51$

Fig. 1: Cadence schematics of the CMOS inverters for the simulated ring oscillators with gate dimensions and load capacitance calculated for an intended N.

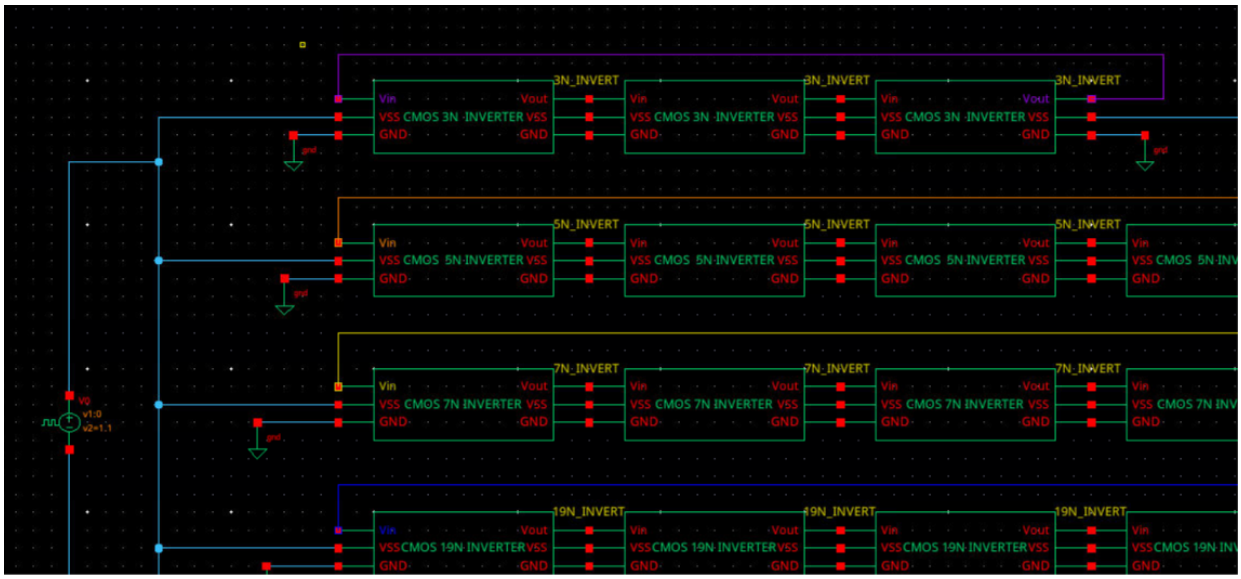


Fig. 2: Compared ring oscillator schematic using 1xN arrays of single inverters.

After connecting the source and ground to each oscillator and applying a stepped input voltage to generate a transient before reaching steady state, the devices were simulated, and their transients are plotted below in Fig. 3. It is seen that larger chains reach steady state faster; this is due to the fact that the total delay time from the first inverter to the last should be the same for each number of N in an ideal design, but since each inverter stage is works as an amplifier, the signal reaches its maximum

faster if there are more inverters in the chain. At steady state, it can be seen in Fig. 4 that a higher N produces a purer square wave; this is because the loop delay, $t_{loop} = n\,t_p$ is ideally the same for each design, so the higher N designs use faster inverters. A ring of more, faster stages spends about $1/(2n)$ of each period switching, which equivalently pushes each stage's bandwidth far enough above the fundamental to pass the harmonics that make up a square wave.

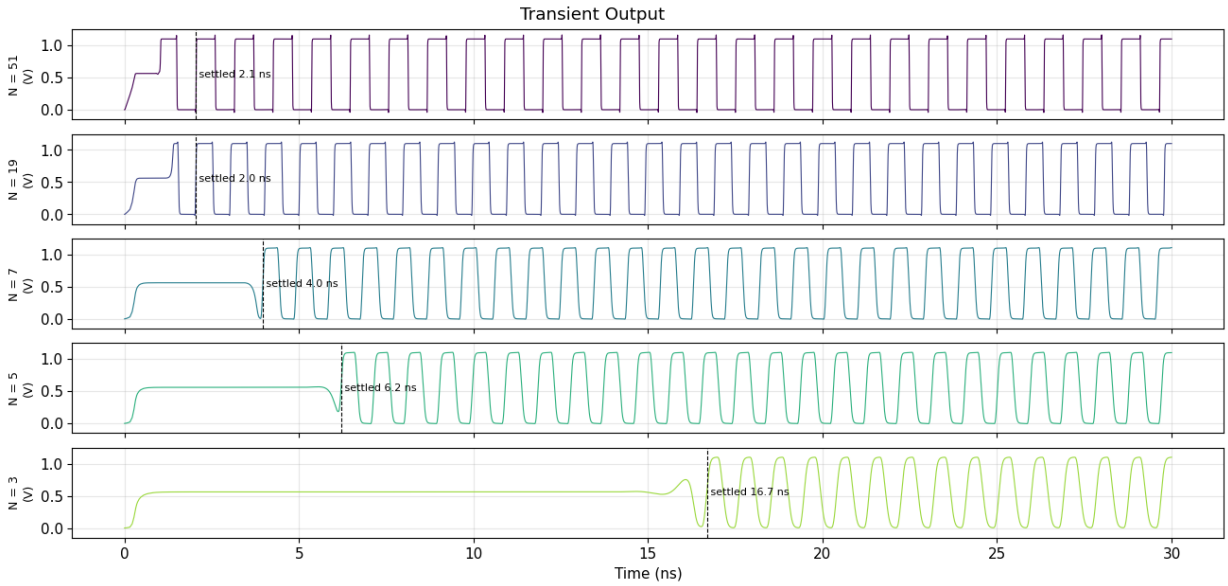


Fig. 3: Transient behavior of all 5 designs with marked settling time.

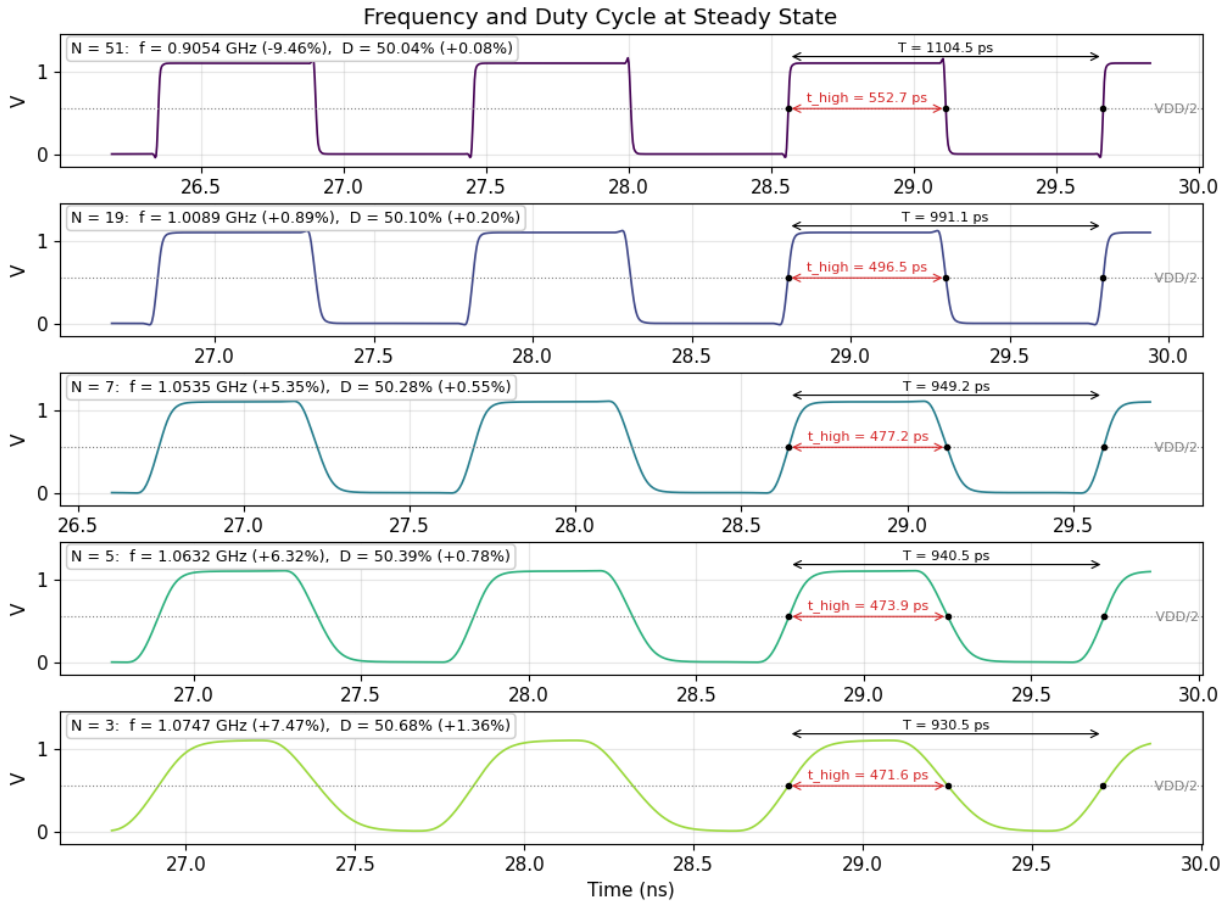


Fig. 4: Duty cycle and steady state frequency calculated from the last 2 simulated periods for each designs.

The duty cycle and frequency of each wave were calculated at steady state, and the error was compared in Fig. 5. It can be seen that the error drops as N increases, but passes the center for frequency. In a more in-depth experiment, it would be important to explore this to determine whether the frequency error varies with N.

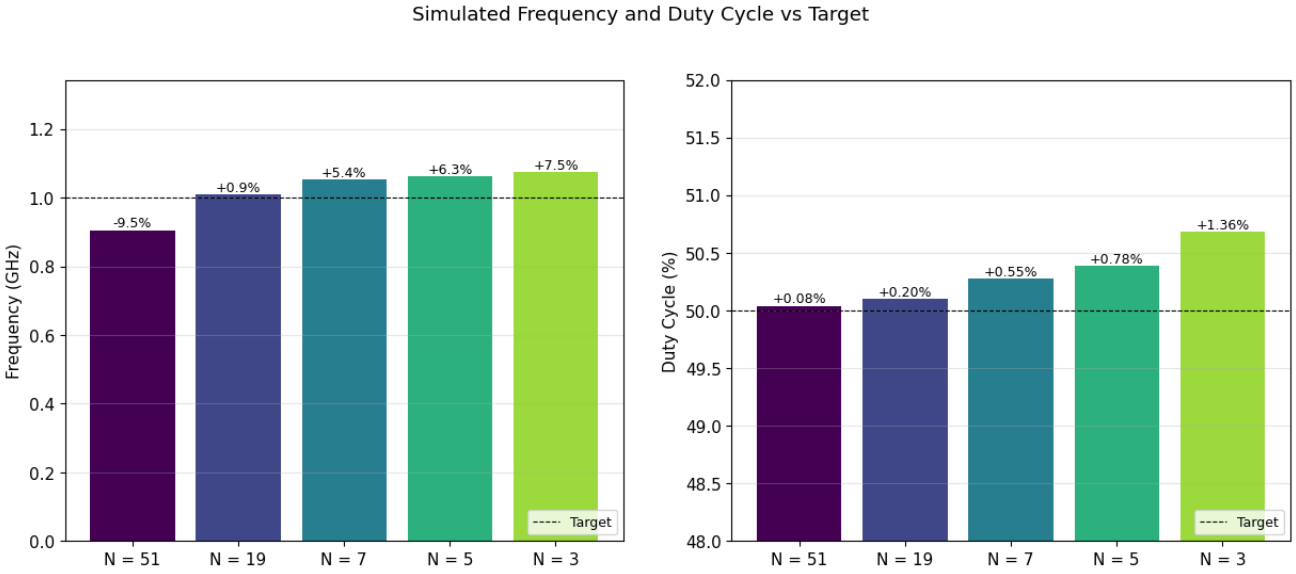


Fig. 5: Error between oscillators of different size relative to target frequency and duty cycle.

IV. Conclusion

The higher N stages show purer square waves, but will take up more space on the die. Sometimes the square wave must be very pure and chip space is less of a concern, in which case large oscillators may be best, but this is not always the case. In this specific experiment, the error was reduced as N grew for duty cycle, but got larger between N=19 and N=51, likely due to a parameter discrepancy whose effect is greater at large N, such as an underestimated device capacitance, which would have compounding effects as N grows. To determine the exact cause, more simulations must be carried out to determine if this is a growing error or a one-off case.